iLearn Practice Exam

12BR · Chapter / Topic 15

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Topic 15: Hardware

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 62

Marks obtained: ..... / 62

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 62.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, mark your work against the official Cambridge mark scheme (see the source list on the last page).

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 This truth table represents a logic circuit.

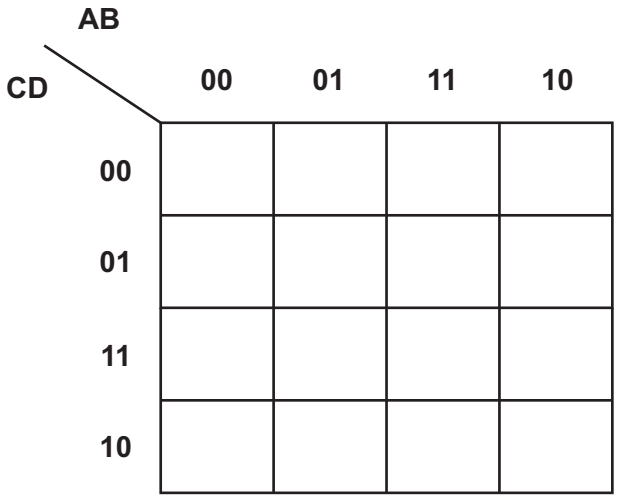
INPUT				OUTPUT
A	**B**	**C**	**D**	**Z**
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

(a) Write the Boolean logic expression that corresponds to the given truth table as the sum-of-products.

Z = ...

..... [2]

(b) (i) Complete the Karnaugh map (K-map) for the given truth table.



[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean logic expression from your answer to part **(b)(ii)** as the simplified sum-of-products.

Z = ...

..... [2]

2 **(a)** Describe the characteristics of massively parallel computers.

.....

.....

.....

.....

.....

..... [3]

(b) State **one** difference and **one** similarity between the SIMD and MISD computer architectures.

difference .....

.....

.....

similarity .....

.....

.....
[2]

3 **(a)** The Karnaugh map (K-map) represents a logic circuit with four inputs.

CD \ AB	00	01	11	10
00	0	1	1	0
01	0	1	0	0
11	0	1	0	0
10	0	1	1	0

(i) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(ii) Write the Boolean expression from your answer to **6(a)(i)** as a simplified sum-of-products. Do **not** carry out any further simplification.

.....

..... [2]

(iii) Write the Boolean expression from your answer to **6(a)(ii)** in its simplest form by using Boolean algebra to simplify it.

..... [1]

(b) Simplify the following expression using De Morgan's laws and Boolean algebra.

Show all the stages in your simplification.

$\mathbf{X = \overline{A + B + C} + \overline{B + C}}$

.....

.....

.....

.....

.....

..... [3]

4 Explain what is meant by the term virtual machine.

.....

.....

.....

.....

.....

.....

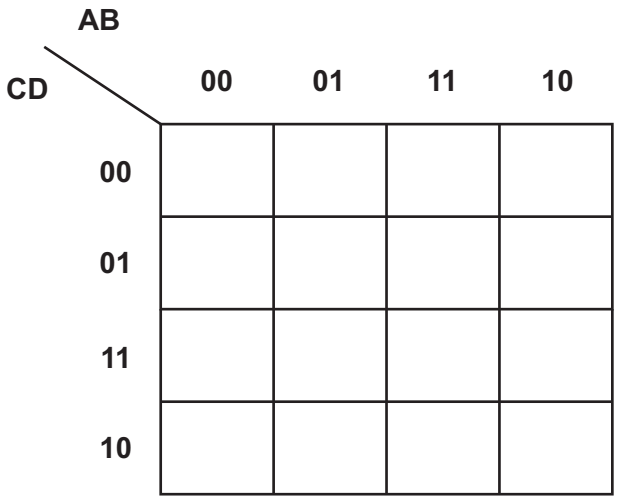
.....

..... [4]

5 This truth table represents a logic circuit.

INPUT				OUTPUT
A	**B**	**C**	**D**	**Z**
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

(a) (i) Complete the Karnaugh map (K-map) for the given truth table.



[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to **7(a)(ii)** as a simplified sum-of-products.

.....

..... [1]

(b) Simplify the following expression using De Morgan's laws and Boolean algebra.

Show all the stages in your simplification.

$$X = \overline{\overline{A}.B.C.\overline{D}} + \overline{\overline{A}.D}$$

.....

.....

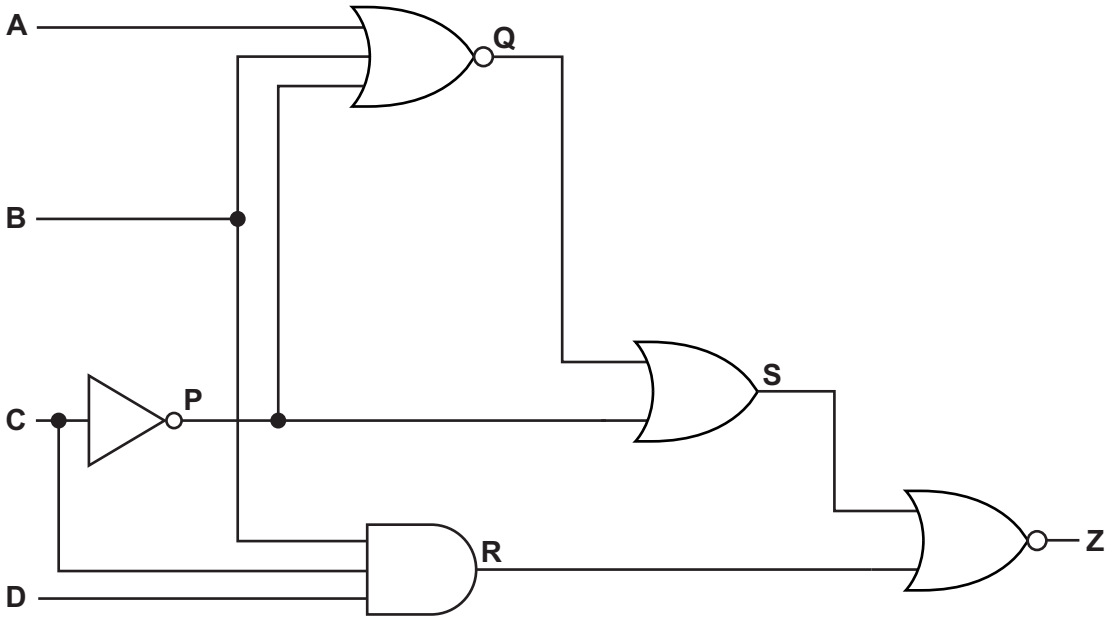
.....

.....

.....

..... [3]

6 The diagram shows a logic circuit.



(a) Complete the truth table for the given logic circuit.
Show your working.

				Working space				
A	**B**	**C**	**D**	**P**	**Q**	**R**	**S**	**Z**
0	0	0	0					
0	0	0	1					
0	0	1	0					
0	0	1	1					
0	1	0	0					
0	1	0	1					
0	1	1	0					
0	1	1	1					
1	0	0	0					
1	0	0	1					
1	0	1	0					
1	0	1	1					
1	1	0	0					
1	1	0	1					
1	1	1	0					
1	1	1	1					

(b) Write the Boolean logic expression that corresponds to the given logic circuit as the sum-of-products.

Z = ...

..... [1]

(c) Use Boolean algebra including De Morgan's laws to simplify the following expression. Show all working.

$$(\overline{\mathbf{A} + \mathbf{B}}) \mathbf{.} (\overline{\mathbf{A} \mathbf{.} \mathbf{B}} + \overline{\mathbf{B} \mathbf{.} \mathbf{C}})$$

Working .....

.....

.....

.....

.....

.....

Simplified expression .....

.....

[4]

7 A virtual machine may be used to emulate computer hardware using software.

Outline **two** benefits and **two** limitations of using a virtual machine.

benefit 1 .....

.....

benefit 2 .....

.....

limitation 1 .....

.....

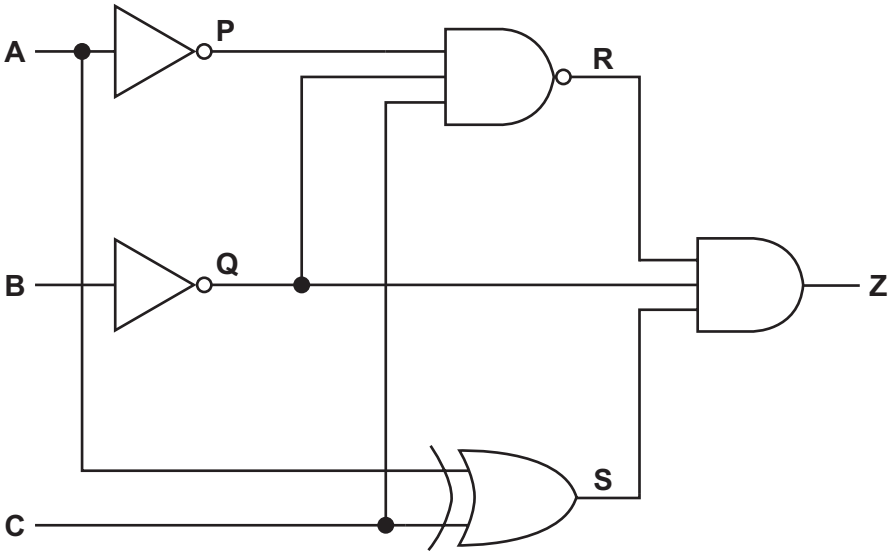
limitation 2 .....

.....

[4]

—

8 **(a)** The diagram shows a logic circuit.



Complete the truth table for the given logic circuit.
Show your working.

			working space				
A	**B**	**C**	**P**	**Q**	**R**	**S**	**Z**
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

[3]

(b) Describe **one** benefit of using a Karnaugh map (K-map) for solving logic problems.

.....

..... [1]

(c) The Karnaugh map (K-map) represents a logic circuit with four inputs.

CD \ AB	00	01	11	10
00	0	0	1	0
01	0	1	1	0
11	1	1	1	1
10	0	0	1	0

(i) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [3]

(ii) Write the Boolean expression from your answer to **6(c)(i)** as a simplified sum-of-products. Do **not** carry out any further simplification.

.....

..... [2]

9 This truth table represents a logic circuit.

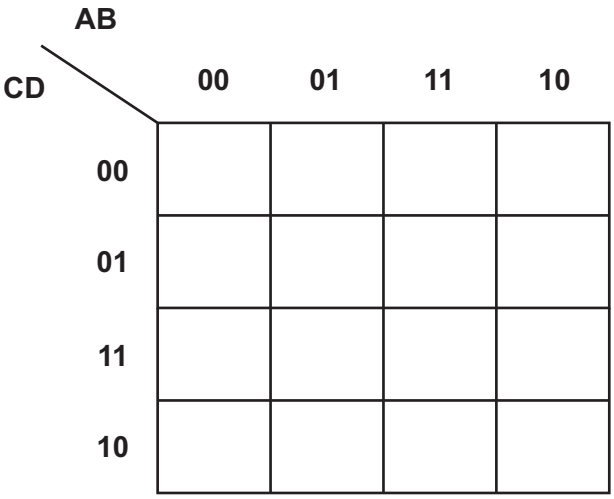
INPUT				OUTPUT
A	**B**	**C**	**D**	**Z**
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

(a) Write the Boolean logic expression that corresponds to the given truth table as the sum-of-products.

Z = ...

..... [2]

(b) (i) Complete the Karnaugh map (K-map) for the given truth table.



[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean logic expression from your answer to part **(b)(ii)** as the simplified sum-of-products.

Z = ...

..... [2]

END OF QUESTION PAPER

Source of each question (for marking)

Find the matching official Cambridge mark scheme for the series/paper shown, to mark your answers.

Q	Past paper	Original question	Marks
1	Oct/Nov 2025 Paper 31	Q5 (a),(b)	8
2	May/June 2026 Paper 33	Q4 (a),(b)	5
3	May/June 2026 Paper 33	Q6 (a),(b)	8
4	May/June 2026 Paper 32	Q5	4
5	May/June 2026 Paper 32	Q7 (a),(b)	8
6	Oct/Nov 2025 Paper 32	Q7 (a),(b),(c)	8
7	May/June 2026 Paper 31	Q5	4
8	May/June 2026 Paper 31	Q6 (a),(b),(c)	9
9	Oct/Nov 2025 Paper 33	Q3 (a),(b)	8

Total: 62 marks